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Map-Free Causal GPS Trajectory Denoising with Rectified Flow

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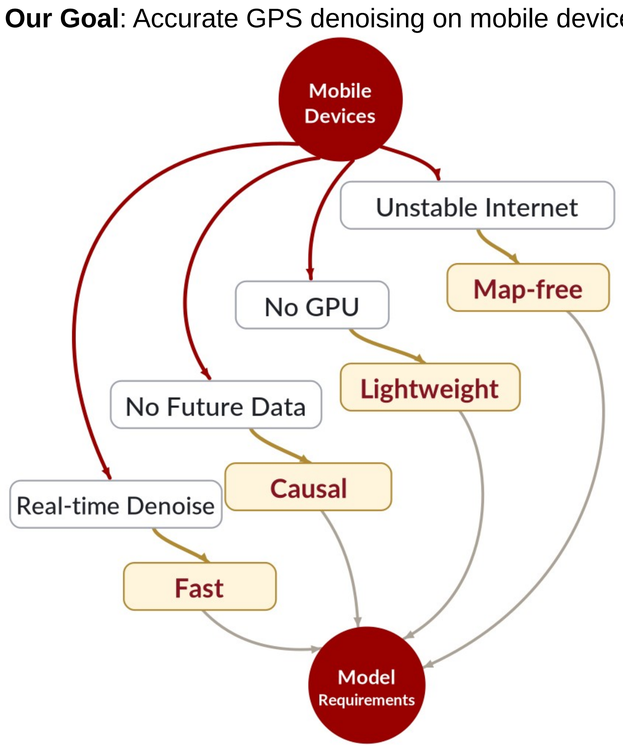
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Motivation

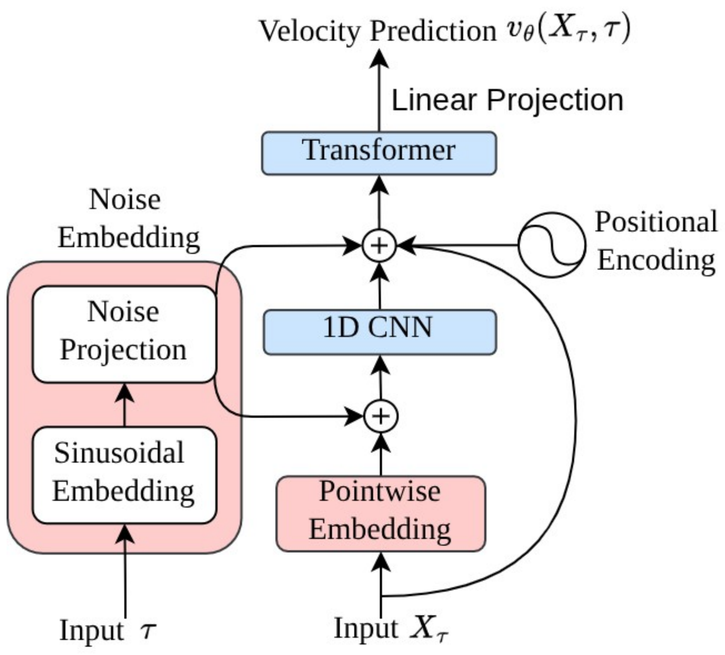
GPS Noise: Degrades positioning accuracy and mobility applications.

Existing Methods: Limited by residual errors, map dependency, or non-causal processing.

Our Goal: Accurate GPS denoising on mobile devices.



Model Architecture



Rectified Flow

(a) Linear Interpolation

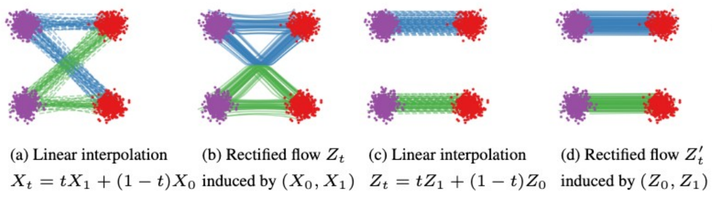
$$X_\tau = (1-\tau)X_0 + \tau X_1, \quad \tau \in [0,1]$$

(b) Rectified Flow — Velocity Learning

$$v_\theta(X_\tau, \tau) \approx X_1 - X_0$$

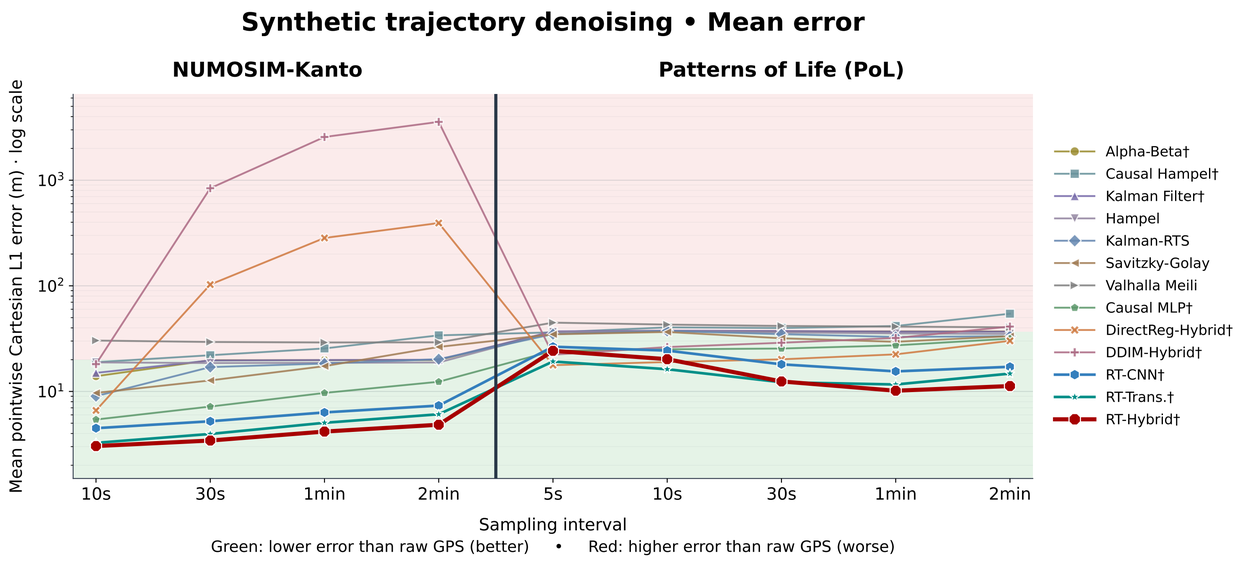
One-Step Inference

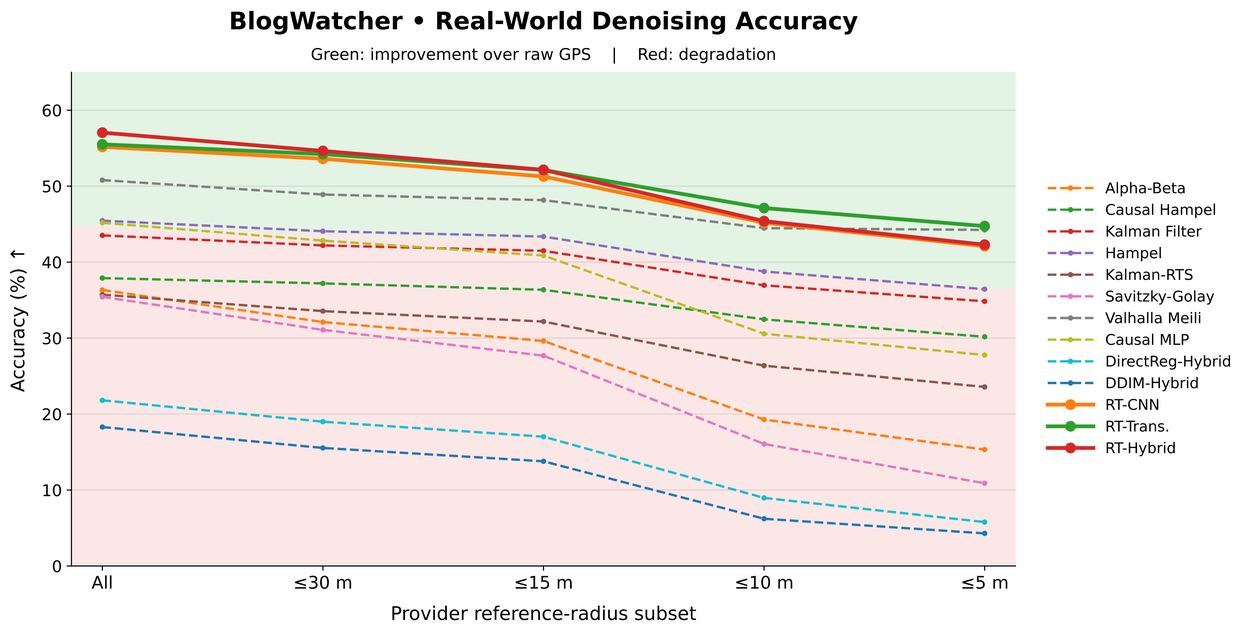
$$\hat{X}_0 = X_1 - v_\theta(X_1, 1)$$

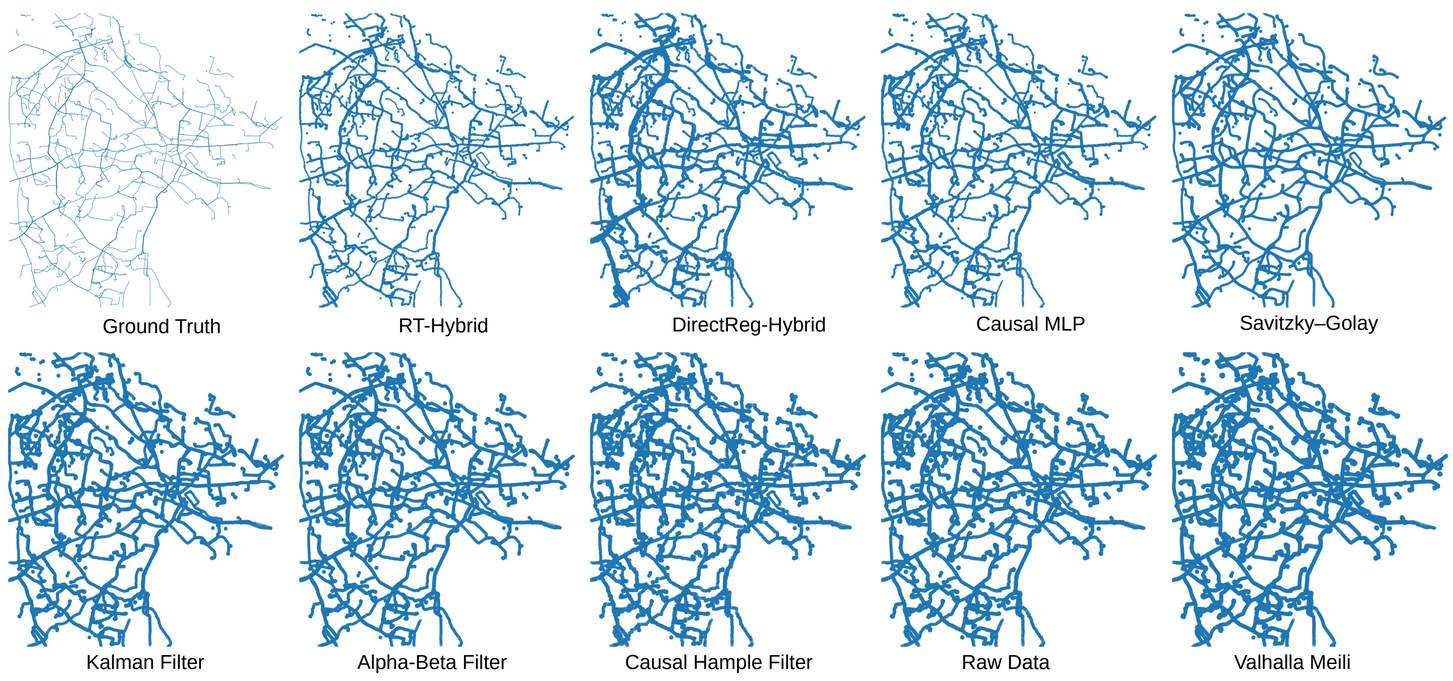


(a) Linear interpolation $X_t = tX_1 + (1-t)X_0$ (b) Rectified flow Z_t induced by (X_0, X_1) (c) Linear interpolation $Z_t = tZ_1 + (1-t)Z_0$ (d) Rectified flow Z'_t induced by (Z_0, Z_1)

Experimental Result

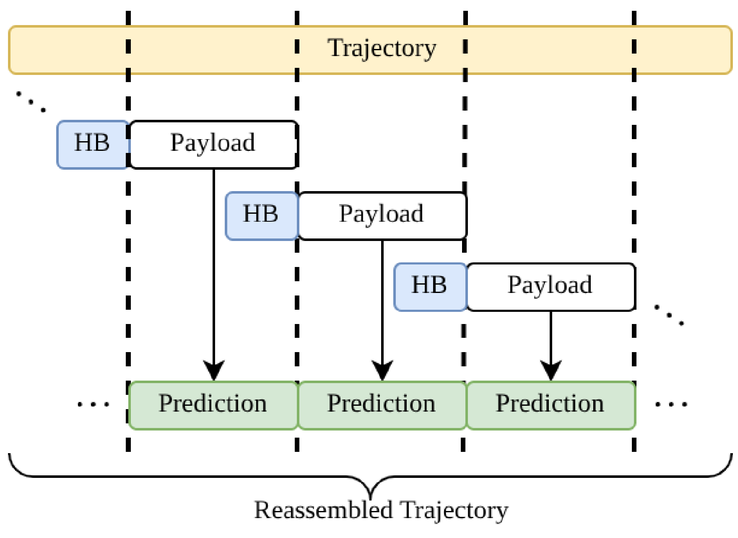






Pipelines

Head Buckle (HB): Reuses historical context to reduce errors at chunk boundaries.



Conclusion

Causal & Map-Free: GPS denoising without future observations, road maps, or external services.

One-Step Correction: Rectified flow learns noise displacement for direct trajectory correction.

Accuracy: Outperforms classical filters on simulated and real-world datasets.

Generalization & Robustness: Zero-shot transfer across noise processes and sampling intervals; more stable than direct regression.

Mobile Efficiency: ~1M parameters; CPU-only Pixel 4a inference in 11–124 ms per 256-point window.

Research Paper | Extended Study | GitHub